

$$\text{NL}_{\mathbf{k}_s \rightarrow 0} \left(\begin{array}{c} \square \\ \delta h_{\mathbf{k}_1} \\ \diagdown \\ \bigcirc \\ \diagup \\ \delta h_{\mathbf{k}_2} \\ \square \end{array} \begin{array}{c} \text{---} A_{\mathbf{q}}^{\mu} \text{---} \\ \text{---} A_{\mathbf{q}+\mathbf{k}_s}^{\mu} \text{---} \end{array} \begin{array}{c} \diagup \\ \delta h_{\mathbf{k}_3} \\ \square \\ \diagdown \\ \delta h_{\mathbf{k}_4} \\ \square \end{array} \right) = \begin{array}{c} \square \\ \diagdown \\ \bigotimes \\ \diagup \\ \square \end{array} \times \begin{array}{c} \bullet \text{---} \text{---} \bullet \end{array} \times \begin{array}{c} \square \\ \diagup \\ \bigotimes \\ \diagdown \\ \square \end{array}$$

The diagram illustrates a factorization property of a nonlinear interaction in the limit $\mathbf{k}_s \rightarrow 0$. On the left, a large bracket labeled $\text{NL}_{\mathbf{k}_s \rightarrow 0}$ encloses a diagram with four external legs (squares) and a central loop. The left vertex is a white circle with two incoming legs labeled $\delta h_{\mathbf{k}_1}$ and $\delta h_{\mathbf{k}_2}$. The right vertex is also a white circle with two outgoing legs labeled $\delta h_{\mathbf{k}_3}$ and $\delta h_{\mathbf{k}_4}$. The loop consists of two wavy blue lines, with the top arc labeled $A_{\mathbf{q}}^{\mu}$ and the bottom arc labeled $A_{\mathbf{q}+\mathbf{k}_s}^{\mu}$. This is set equal to the product of three separate diagrams: a vertex $\bigotimes$ with two incoming legs, a propagator consisting of two wavy blue lines with black dots at the ends, and a vertex $\bigotimes$ with two outgoing legs.